

Serin Baek

CURRICULUM VITAE

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EDUCATION

University College London

MSci Pharmacology

First Class Honors (on track)

Awards: D.H. Jenkinson Prize, Dean's List

2023 - present

RESEARCH POSITIONS

Schiavo Group, Queen Square Institute of Neurology

Supervised by Professor Giampietro Schiavo and Kai Sun

Investigated the structural assembly of cytoplasmic dynein-1 under conditions of LIC1/LIC2 serine phosphorylation and G-domain overexpression using Co-IP and Western Blot competitive binding assays.

September 2026

Fitzgerald's Group, Early Cancer Institute, University of Cambridge

Supervised by Professor Rebecca Fitzgerald, Dr. Ahsen Ustaoglu and Timothy Somerset

Evaluated non-endoscopic capsule sponge assays and liquid biopsies for the early detection of oesophageal adenocarcinoma. Performed SIMOA and bioinformatic methylation analyses (EM-Seq vs. 5-Base) to identify and validate candidate protein (FABP2 and REG4) and epigenetic biomarkers. Compared the efficiency of 5-Base sequencing and EM-Seq in capturing methylation signal as an epigenetic biomarker for Barrett's.

July 2026 - August 2026

Moss Group, UCL

Supervised by Dr. Guy Moss and Dr. Thomas Blacker

Investigated gap-junctional coupling between airway epithelial cells and how CFTR-expressing cells may electrically support their non-expressing neighbours. Built a resistor-network computational model in R to fit published datasets and conducted FLIM using a genetically encoded voltage indicator (JEDI-2P) in HEK293T cells to measure membrane potential.

June 2026 - September 2026

Sihra, UCL

Supervised by Professor Talvinder Sihra

Analysed 82 peer reviewed studies, writing a 5000-word review on *Phosphorylation Dependent Modulation of Presynaptic Plasticity*, characterising a theoretical presynaptic multi-phosphocode, mapping how the temporal balance between competing kinase cascades encodes the direction and magnitude of synaptic plasticity.

September 2025 - June 2026

Centre for Cardiovascular and Metabolic Neuroscience (CCMN), UCL

Supervised by Professor Alexander Gourine and Dr. Alice Braga

Explored how astrocytic A2B receptor activation regulates brain energy metabolism and lactate release to support synaptic plasticity. Performed cryostat sectioning and immunostaining on murine brains to evaluate metabolic signalling pathways linked to cognitive health and neurodegeneration.

January 2025 - April 2025

AWARDS AND FUNDING

Dean's List, UCL

Awarded in recognition of exceptional academic performance in the Faculty of Life Sciences (top 5%).

August, 2026

Early Cancer Institute Summer Research sponsored by Moderna

Fully funded 8-week summer studentship at the Early Cancer Institute at the University of Cambridge.

July, 2026

Cystic Fibrosis Trust Summer Studentship Scheme, Cystic Fibrosis Trust UK

Fully funded 7-week summer research on bioelectric profiling of airway epithelial cell types in cystic fibrosis.

June, 2026

D.H. Jenkinson Prize, UCL

Awarded to the top-performing student in the MSci/BSc Pharmacology cohort for best overall first-year academic performance (81% average).

September, 2024

Presidential Science Scholar (Top 20 Nationwide), South Korea

2023-2027

Awarded the most prestigious scholarship by the South Korean government, fully funding undergraduate studies abroad (\$200,000 USD), with no bond requirement.

RESEARCH TALKS

Serin Baek, Thomas Blacker, and Guy Moss. Bioelectric Profiling and Ion Transport in Airway Epithelial Cells. (2026). Cystic Fibrosis Trust, September 23, London, United Kingdom.

Serin Baek, Ahsen Ustaoglu, Timothy Somerset, and Rebecca Fitzgerald. Translating Non-endoscopic Assays into Novel Diagnostics for Early Detection of Oesophageal Adenocarcinoma. (2026). Early Cancer Institute, August 27, Cambridge, United Kingdom.

WORK EXPERIENCE

Operational Excellence Summer Intern, AstraZeneca UK

June 2025 - September 2025

Designed a Power BI dashboard for the UK-Sweden team to track KPIs, developed Excel VBA automation for SAP processes, and analysed 3,000+ Supplier Quality Risk Assessments to identify risk trends and support senior-level decision making. Received the You Made the Difference Award and a Gold certificate on Thriving in the age of AI.

LEADERSHIP EXPERIENCE

Student Ambassador, UK Council for International Student Affairs (UKCISA)

December 2025 - present

Student ambassador of UK's national advisory board, providing lived-experience insight to improve support for international students in the UK. Invited to global education and policy forums as a guest speaker, including The PIE Live and UKCISA Annual Conference.

Room Monitor Supervisor, UCL

October 2024 - present

Oversee all room monitors of UCL's administrative office to ensure operational readiness. Accumulated 600+ hours of early-morning room monitoring (8AM start), demonstrating consistency and diligence.

Student President, East-West International School, Cambodia

August 2021 - Dec 2022

Initiated a volunteering partnership with Cambodia Children's Fund, designed 200+ student report card templates and developed Excel-based class timetables in collaboration with fellow interns.

Ambassador of Yale Young Global Scholars (YYGS), Cambodia

November 2021 - June 2022

Organised YYGS information sessions with the YYGS director to bring in the opportunity to learn for students in Cambodia.

VOLUNTEERING EXPERIENCE

Homeless Outreach, Euston Church

October 2025 – May 2026

Baked desserts and spoke to homeless people every week, acting as the listening ears while gaining insight into how social vulnerability increases health challenges.

Volunteering and Welfare Officer, Music Society, UCL

September 2024 - June 2025

Organised 4+ volunteering concerts at hospitals and schools, coordinating logistics and performance schedules with UCL Student Union and musicians.

Mobile Clinic Medical Volunteer, One Peninsula Medical Union, Cambodia

July 2024

Supported clinicians in safe and effective dosing decisions, interpreting (Khmer) between doctors and patients to ensure clear communication.

TEACHING EXPERIENCE

East-West International School

September 2022 – December 2022

Assistant teacher for A-Level Further Mathematics, as a valedictorian with an A* in A-Level Maths.

SKILLS

Programming: R, Python

Electrophysiology & Wet Lab: Western Blot, molecular cloning, dose-response assays, patch clamp, mammalian/neuronal cell culture, transfection, fluorescence microscopy, FLIM, SIMOA, ELISA, BCA Protein Assay

Imaging & Analysis: Andor Solis, Fiji (ImageJ), GraphPad Prism, HEKA Pulse, Excel